9. Least squares

- least squares problem
- solution and normal equations
- multi-objective least squares
- control
- estimation and inversion

Least squares problem

- consider $Ax = b$ where A is an $m \times n$ matrix and b is an m-vector
- in most applications, system is overdetermined $m > n$ without a solution

Least squares problem: choose x that minimizes the residual norm $r = Ax - b$:

$$\text{minimize} \quad \|Ax - b\|^2 = \sum_{i=1}^{m} \left(\sum_{j=1}^{n} A_{ij} x_j - b_i \right)^2$$

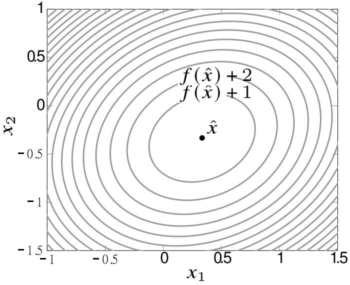
- x is *variable*, A, b are called *data*, $\|Ax - b\|^2$ is the *objective* function
- also called *regression* (in data fitting and statistics context)
- $\hat{x}$ is a *solution* of the least squares problem if it gives the smallest objective:

$$\|A\hat{x} - b\|^2 \leq \|Ax - b\|^2 \quad \text{for any } n\text{-vector } x$$

  - $\hat{x}$ also called *least squares approximate solution of* $Ax = b$
  - if $\hat{r} = A\hat{x} - b = 0$, then $\hat{x}$ solves linear equation $Ax = b$

Example

$$A = \begin{bmatrix} 2 & 0 \\ -1 & 1 \\ 0 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$



- $Ax = b$ has no solution
- least squares problem:

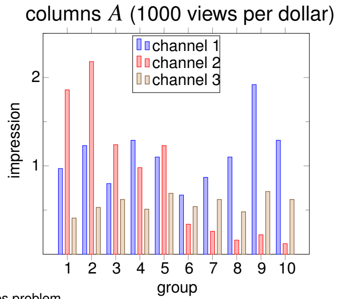
$$\text{minimize} \quad \|Ax - b\|^2 = (2x_1 - 1)^2 + (-x_1 + x_2)^2 + (2x_2 + 1)^2$$

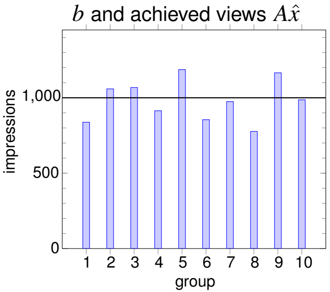
- least squares solution is $\hat{x} = (1/3, -1/3)$
- $\|A\hat{x} - b\|^2 = 2/3$ is smallest posible value of $\|Ax - b\|^2$

Example: advertising purchases

- m demographics groups (audiences), n advertising channels
- b_i is target number of views or impressions for group i
- A_{ij} is # views in group i per dollar spent on ads in channel j
- x_j is amount of advertising purchased in channel j
- $(Ax)_i$ is total number of views in group i
- least squares problem: minimize $\|Ax - b\|^2$ (ignoring $x \geq 0$ and budget)

Example: $m = 10$, $n = 3$, $b = (10^3)\,\mathbf{1}$, $\hat{x} = (62, 100, 1443)$

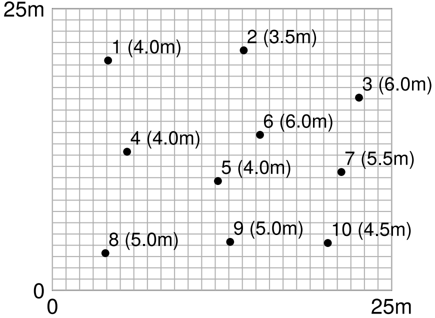




Example: illumination

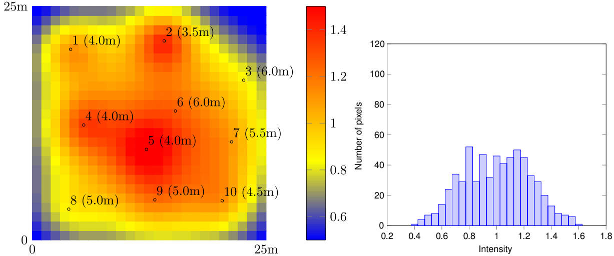
- n lamps illuminate an area divided in m regions
- b_i is target illumination level at region i
- x_j is power of lamp j
- A_{ij} is illumination in region i if lamp j is on with power 1, other lamps are off
- $(Ax)_i$ is illumination level at region i

Example: lamp positions and heights with $m = 25 \times 25$, $n = 10$

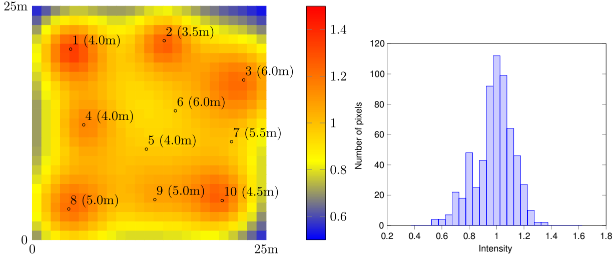


Illumination

equal lamp powers $(x = \mathbf{1})$



least squares solution $\hat{x}$, with $b = \mathbf{1}$



Outline

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Least squares solution

$$\text{minimize} \quad \|Ax - b\|^2$$

Normal equations: a solution $\hat{x}$ must satisfy the *normal equations:*

$$A^T A\hat{x} = A^T b$$

if A has *linearly independent columns*, then the solution is unique

$$\hat{x} = (A^T A)^{-1} A^T b = A^\dagger b$$

- $A^\dagger = (A^T A)^{-1} A^T$ is the psuedo-inverse of A, which is also a left inverse
- $\hat{x} = A^\dagger b$ solves the linear equation $Ax = b$ if it has a solution
- if $Ax = b$ does not have a solution, then $A\hat{x} \neq b$

Proof using algebra

suppose $\hat{x}$ satisfies the normal equations $A^T(A\hat{x} - b) = 0$, then for any n-vector x

$$\begin{aligned}
\|Ax - b\|^2 &= \|(Ax - A\hat{x}) + (A\hat{x} - b)\|^2 \\
&= \|A(x - \hat{x})\|^2 + \|A\hat{x} - b\|^2 + 2(A(x - \hat{x}))^T(A\hat{x} - b) \\
&= \|A(x - \hat{x})\|^2 + \|A\hat{x} - b\|^2 + 2(x - \hat{x})^T A^T(A\hat{x} - b) \\
&= \|A(x - \hat{x})\|^2 + \|A\hat{x} - b\|^2
\end{aligned}$$

- hence for any x, $\|Ax - b\|^2 \geq \|A\hat{x} - b\|^2$
- if A has linearly independent columns, then the solution is unique:

$$\|Ax - b\|^2 > \|A\hat{x} - b\|^2$$

  this is because $\|A(x - \hat{x})\|^2 = 0 \Rightarrow A(x - \hat{x}) = 0 \Rightarrow x = \hat{x}$

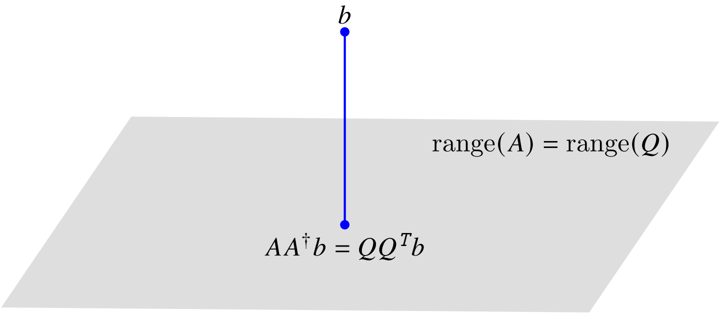
Geometric interpretation: projection on range

- using QR factorization $A = QR$, we have $A^\dagger = R^{-1}Q^T$ and

$$AA^\dagger = QRR^{-1}Q^T = QQ^T$$

note that $AA^\dagger$ and is different from $A^\dagger A = I$

- $AA^\dagger b = QQ^T b$ is the projection of b onto $\mathrm{range}(Q) = \mathrm{range}(A)$ (page 5.6)



Geometric interpretation: orthogonality condition

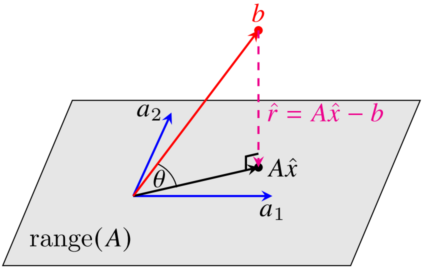
let $a_1, \ldots, a_n$ denote columns of A, then $\|Ax - b\|^2 = \|(x_1 a_1 + \cdots + x_n a_n) - b\|^2$

- $A\hat{x}$ is the vector in range(A) = span$(a_1, \ldots, a_n)$ closest to b:

  it is the (orthogonal) projection of b on range(A)

- geometrically, $\hat{r} = A\hat{x} - b$ is orthogonal to range(A) = span$(a_1, \ldots, a_n)$:

$$A^T \hat{r} = 0 \quad \Longrightarrow \quad A^T A \hat{x} = A^T b$$



Example: concrete mixture

given the weight percentages (per lbs) of two concrete types:

material	**type 1 (%)**	**type 2 (%)**
cement	30%	10%
gravel	40%	20%
sand	30%	70%

- determine weights of two mixtures to achieve a target mixture of:

  5.0 lbs cement, 3.0 lbs gravel, and 4.0 lbs sand

- goal: minimize the difference to reach these targets

Example: concrete mixture

- letting x_1 and x_2 to be the weights of concrete of the first and second types
- the problem can be formulated as the least squares problem:

$$\text{minimize} \quad \left\| \begin{bmatrix} 0.3 & 0.1 \\ 0.4 & 0.2 \\ 0.3 & 0.7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 5 \\ 3 \\ 4 \end{bmatrix} \right\|^2 = \|Ax - b\|^2$$

  where $x = (x_1, x_2)$

- since the columns of A are linearly independent, the solution is

$$\hat{x} = (A^T A)^{-1} A^T b = \begin{bmatrix} 10.6 \\ 0.961 \end{bmatrix}$$

Solution using QR factorization method

using QR factorization $A = QR$, we have

$$\begin{aligned}\hat{x} = (A^T A)^{-1} A^T b &= \left((QR)^T (QR)\right)^{-1} (QR)^T b \\ &= (R^T Q^T Q R)^{-1} R^T Q^T b \\ &= R^{-1} Q^T b\end{aligned}$$

- identical formula for solving $Ax = b$ for square invertible A
- here $\hat{x}$ gives least squares approximate solution to $Ax = b$

Algorithm

1. compute QR factorization $A = QR$ ($2mn^2$ flops if A is $m \times n$)
2. matrix-vector product $Q^T b$ ($2mn$ flops)
3. solve $Rx = Q^T b$ by back substitution (n^2 flops)

Complexity: $\sim 2mn^2$ flops

Example

$$A = \begin{bmatrix} 3 & -6 \\ 4 & -8 \\ 0 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} -1 \\ 7 \\ 2 \end{bmatrix}$$

1. QR factorization: $A = QR$ with

$$Q = \begin{bmatrix} 3/5 & 0 \\ 4/5 & 0 \\ 0 & 1 \end{bmatrix}, \quad R = \begin{bmatrix} 5 & -10 \\ 0 & 1 \end{bmatrix}$$

2. calculate $d = Q^T b = (5, 2)$

3. solve $Rx = d$

$$\begin{bmatrix} 5 & -10 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

solution is $x_1 = 5, x_2 = 2$

Solution using normal equations

---

given $m \times n$ matrix A with linearly independent columns and n-vector b

1. form $B = A^T A$ and $y = A^T b$
2. compute the Cholesky factorization $B = R^T R$ (R is lower triangular)
3. solve $R^T z = y$ for z using forward substitution
4. solve $Rx = z$ for x using back substitution

Complexity: approximately $mn^2 + n^3/3$ (flops)

- step 1 costs mn^2
- step 2 is approximately $n^3/3$ flops
- steps 3 and 4 cost order n^2 flops
- when $m \gg n$, the main cost becomes in forming the matrix $B = A^T A$

Comparison of the two methods

Complexity

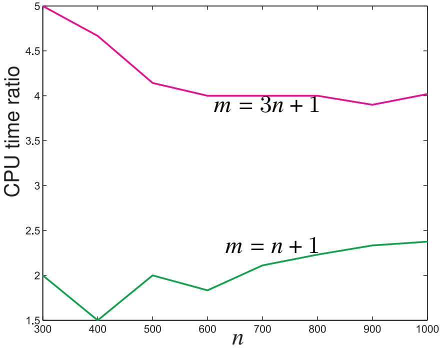
- Cholesky method: $mn^2 + (1/3)n^3$ flops
- QR method: $2mn^2$ flops
- Cholesky method is faster by a factor of two (if $m \gg n$)

Numerical stability: QR factorization method is more stable

- QR method computes R without "squaring" A (*i.e.*, forming A^TA)
- this is important when the columns of A are "almost" linearly dependent

Example

- randomly create A and a vector b
- plot **ratio** of CPU times for using QR way over normal equations way



- normal equations method is more efficient

Code

```
for n = 300:100:1000
% fill a rectangular matrix A and a vector b with random numbers
m = n+1; % or m= 3*n+1
A = randn(m,n); b = randn(m,1);
% solve and find execution times; first, Matlab way using QR
t0 = cputime;
xqr = A\b;
temp = cputime;
tqr(n/100-2) = temp - t0;
% next use normal equations
t0 = temp;
B = A'* A; y = A'*b;
xne = B\y;
temp = cputime;
tne(n/100-2) = temp - t0;
end
ratio = tqr./tne;
plot(300:100:1000,ratio)
```

Solving the normal equations

- last example shows direct method is faster
- however, QR method is more stable as illustrated next

Example: a 3×2 matrix with "almost linearly dependent" columns

$$A = \begin{bmatrix} 1 & -1 \\ 0 & 10^{-5} \\ 0 & 0 \end{bmatrix}, \quad b = \begin{bmatrix} 0 \\ 10^{-5} \\ 1 \end{bmatrix}$$

we round intermediate results to 8 significant decimal digits

Method 1: form Gram matrix A^TA and solve normal equations

$$A^TA = \begin{bmatrix} 1 & -1 \\ -1 & 1+10^{-10} \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \quad A^Tb = \begin{bmatrix} 0 \\ 10^{-10} \end{bmatrix}$$

after rounding, the Gram matrix is singular; hence method fails

Method 2: QR factorization of A is

$$Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad R = \begin{bmatrix} 1 & -1 \\ 0 & 10^{-5} \end{bmatrix}$$

rounding does not change any values (in this example)

- problem with method 1 occurs when forming Gram matrix A^TA
- QR factorization method is more stable because it avoids forming A^TA

Standard methods for solving the linear least squares

Normal equations (Cholesky)

- fast, simple, intuitive
- can be unstable when columns of A are "almost" linearly dependent

QR factorization

- this is the "standard" approach (*e.g.*, in MATLAB `A/b`)
- more robust than the normal equations approach
- more computationally expensive than the normal equations approach if $m \gg n$

Singular value decomposition (SVD) (more on this later in course)

- used mostly when columns of A are (almost) dependent
- very robust but more expensive than QR approach

Matrix least squares

$$\text{minimize} \quad \|AX - B\|_F^2$$

- variable is the $n \times k$ matrix $X = [x_1 \ \cdots \ x_k]$
- A is an $m \times n$ matrix and B is an $m \times k$ matrix
- decouples into a set of k ordinary least squares since

$$\|AX - B\|_F^2 = \|Ax_1 - b_1\|^2 + \cdots + \|Ax_k - b_k\|^2$$

  where x_j is the jth column of X and b_j is the jth column of B

- can choose the columns x_j independently, by minimizing $\|Ax_j - b_j\|^2$
- assuming A has linearly independent columns, the solution is $\hat{x}_j = A^\dagger b_j$ or

$$\hat{X} = A^\dagger B$$

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- least squares problem
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Multi-objective least squares

choose n-vector x so that the following objectives are all small

$$J_1 = \|A_1x - b_1\|^2, \; J_2 = \|A_2x - b_2\|^2, \; \ldots, \; J_k = \|A_kx - b_k\|^2$$

- A_i is an $m_i \times n$ matrix, b_i is an m_i-vector, $i = 1, \ldots, k$
- J_i are the objectives in a multi-objective (multi-criterion) optimization problem

Weighted sum objective: choose x that minimizes weighted average

$$J = \lambda_1 J_1 + \cdots + \lambda_k J_k = \lambda_1 \|A_1x - b_1\|^2 + \cdots + \lambda_k \|A_kx - b_k\|^2$$

- $\lambda_1, \ldots, \lambda_k$ are positive *regularization parameters*
- we typically set $\lambda_1 = 1$, and call J_1 the *primary objective*
- terms $\lambda_2 J_2, \ldots, \lambda_k J_k$ are called *regularization terms*
- λ_i gives how much we care about J_i being small, relative to J_1

Weighted sum solution

write weighted-sum objective as

$$J = \left\| \begin{bmatrix} \sqrt{\lambda_1}\,(A_1 x - b_1) \\ \vdots \\ \sqrt{\lambda_k}\,(A_k x - b_k) \end{bmatrix} \right\|^2 = \|\tilde{A}x - \tilde{b}\|^2$$

with

$$\tilde{A} = \begin{bmatrix} \sqrt{\lambda_1} A_1 \\ \vdots \\ \sqrt{\lambda_k} A_k \end{bmatrix}, \quad \tilde{b} = \begin{bmatrix} \sqrt{\lambda_1} b_1 \\ \vdots \\ \sqrt{\lambda_k} b_k \end{bmatrix}$$

Weighted sum solution: assuming columns of $\tilde{A}$ are linearly independent,

$$\begin{aligned} \hat{x} &= (\tilde{A}^T \tilde{A})^{-1} \tilde{A}^T \tilde{b} \\ &= (\lambda_1 A_1^T A_1 + \cdots + \lambda_k A_k^T A_k)^{-1} (\lambda_1 A_1^T b_1 + \cdots + \lambda_k A_k^T b_k) \end{aligned}$$

(here, A_i can be wide, or have dependent columns)

Optimal trade-off curve

Bi-criterion problem: we let $\hat{x}(\lambda)$ be minimizer of bi-criterion objectives

$$J_1 + \lambda J_2 = \|A_1 x - b_1\|^2 + \lambda \|A_2 x - b_2\|^2$$

Pareto optimal point

- $\hat{x}(\lambda)$ is called *Pareto optimal*
- there is no point z that satisfies

$$J_1(z) < J_1(\hat{x}(\lambda)), \quad J_2(z) < J_2(\hat{x}(\lambda))$$

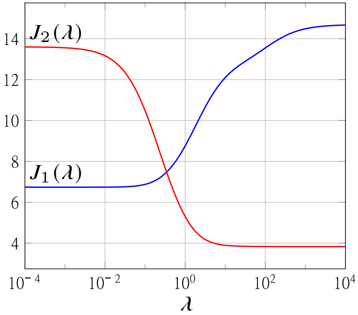
  i.e., no other point beats $\hat{x}$ on both objectives

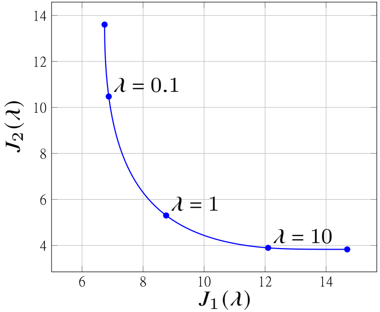
Optimal trade-off curve

$$\big(J_1(\hat{x}(\lambda)), J_2(\hat{x}(\lambda))\big) \quad \text{for} \quad \lambda > 0$$

Example

A_1 and A_2 are both 10×5





- we can achieve a substantial reduction in J_2 with only a small increase in J_1
- weights are typically logarithmically spaced; for N values of $\lambda^{\min} \leq \lambda \leq \lambda^{\max}$:

$$\lambda^{\min}, \quad \theta\lambda^{\min}, \quad \theta^2\lambda^{\min}, \ldots, \quad \theta^{N-1}\lambda^{\min} = \lambda^{\max}$$

with $\theta = (\lambda^{\max}/\lambda^{\min})^{1/(N-1)}$

Tikhonov regularization

the weighted least squares problem

$$\text{minimize} \quad \|Ax - y\|^2 + \lambda\|x\|^2$$

is known as *Tikhonov regularization*

- goal is to make $\|Ax - y\|$ small with x that is not too big
- equivalent to solving
$$(A^T A + \lambda I)x = A^T y$$
- solution is unique (if $\lambda > 0$) even when A has linearly dependent columns

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Control

$$y = Ax + b$$

- n-vector x corresponds to *actions* or *inputs*
- m-vector y corresponds to *results* or *outputs*
- A and b are known (from analytical models, data fitting, ...)
- goal is to choose x, to optimize multiple objectives on x and y

Multi-objective control

- primary objective: $J_1 = \|y - y^{\text{des}}\|^2$, y^{des} is a given desired/target output
- typical secondary objectives:
  - x is small: $J_2 = \|x\|^2$
  - x is not far from a nominal input: $J_2 = \|x - x^{\text{nom}}\|^2$

Optimal input design

Linear dynamical system

$$y(t) = h_0u(t) + h_1u(t-1) + h_2u(t-2) + \cdots + h_tu(0)$$

- output $y(t)$ and input $u(t)$ are scalars
- we assume input $u(t)$ is zero for $t < 0$
- $h_0, h_1, h_2, \ldots$ are the *impulse response coefficients*
- output is convolution of input with impulse response

Optimal input design

- optimization variable is the input sequence $x = (u(0), u(1), \ldots, u(N))$
- goal is to track a desired output using a small and slowly varying input

Input design objectives

$$\text{minimize} \quad J_{\mathrm{t}}(x) + \lambda_{\mathrm{v}} J_{\mathrm{v}}(x) + \lambda_{\mathrm{m}} J_{\mathrm{m}}(x)$$

- primary objective: track desired output y^{des} over an interval $[0, N]$:

$$J_{\mathrm{t}}(x) = \sum_{t=0}^{N} \left(y(t) - y^{\text{des}}(t) \right)^2$$

- secondary objectives: use a small and slowly varying input signal:

$$J_{\mathrm{m}}(x) = \sum_{t=0}^{N} u(t)^2$$

$$J_{\mathrm{v}}(x) = \sum_{t=0}^{N-1} (u(t+1) - u(t))^2$$

Input design objectives

$$J_{\mathrm{t}}(x) + \lambda_{\mathrm{v}} J_{\mathrm{v}}(x) + \lambda_{\mathrm{d}} J_{\mathrm{m}}(x) = \|A_{\mathrm{t}} x - b_{\mathrm{t}}\|^2 + \lambda_{\mathrm{v}} \|Dx\|^2 + \lambda_{\mathrm{d}} \|x\|^2$$

with

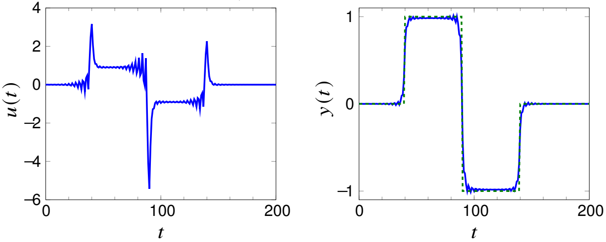
$$A_{\mathrm{t}} = \begin{bmatrix} h_0 & 0 & 0 & \cdots & 0 & 0 \\ h_1 & h_0 & 0 & \cdots & 0 & 0 \\ h_2 & h_1 & h_0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ h_{N-1} & h_{N-2} & h_{N-3} & \cdots & h_0 & 0 \\ h_N & h_{N-1} & h_{N-2} & \cdots & h_1 & h_0 \end{bmatrix}, \quad b_{\mathrm{t}} = \begin{bmatrix} y^{\mathrm{des}}(0) \\ y^{\mathrm{des}}(1) \\ y^{\mathrm{des}}(2) \\ \vdots \\ y^{\mathrm{des}}(N-1) \\ y^{\mathrm{des}}(N) \end{bmatrix}$$

and D is the $N \times (N+1)$ difference matrix

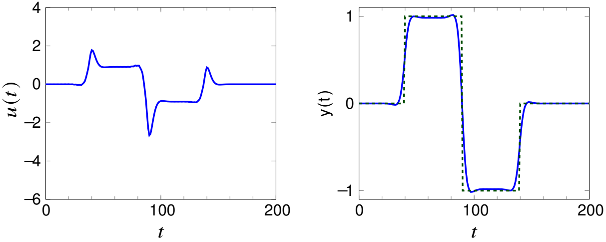
$$D = \begin{bmatrix} -1 & 1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & -1 & 1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -1 & 1 & 0 \\ 0 & 0 & 0 & \cdots & 0 & -1 & 1 \end{bmatrix}$$

Example

$\lambda_{\mathrm{v}} = 0$, small λ_{m}



larger λ_{v} larger λ_{m}



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Estimation (inversion)

measurement model:

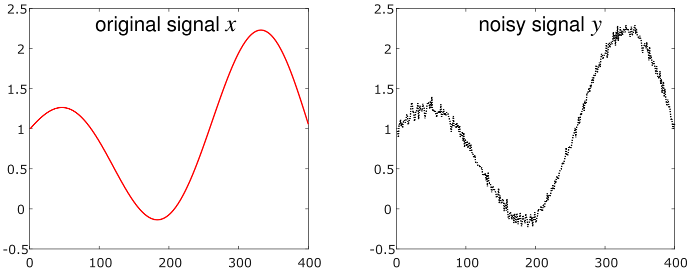
$$y = Ax + v$$

- n-vector x contains parameters we want to estimate
- m-vector y contains the *measurements*
- m-vector v are (unknown) *noises* or *measurement errors*
- $m \times n$ matrix A connects parameters to measurements

Least squares estimation

- we guess x by minimizing $J_1 = \|Ax - y\|^2$
- when v is nonzero or A has dependent columns, we cannot determine x exactly
- in this case, we add other objectives to encode prior information about x
  - x is small: $J_2 = \|x\|^2$
  - x is not far from a nominal input: $J_2 = \|x - x^{\text{nom}}\|^2$

Example: signal denoising



- signal $x = (x_1, x_2, \ldots, x_n)$ can be measured with some additive noise

$$y = x + v \quad \text{where } v \text{ is some noise}$$

- the signal does not vary too much $|x_{i+1} - x_i| << 1$
- given y, we want to find a "good" estimate of x
- least squares formulation: minimize $\quad \|x - y\|^2 + \lambda \sum_{i=1}^{n-1} (x_{i+1} - x_i)^2$

Example: signal denoising

$$\text{minimize} \quad \|x - y\|^2 + \lambda\|Dx\|^2 = \left\| \begin{bmatrix} I \\ \sqrt{\lambda} D \end{bmatrix} x - \begin{bmatrix} y \\ 0 \end{bmatrix} \right\|^2$$

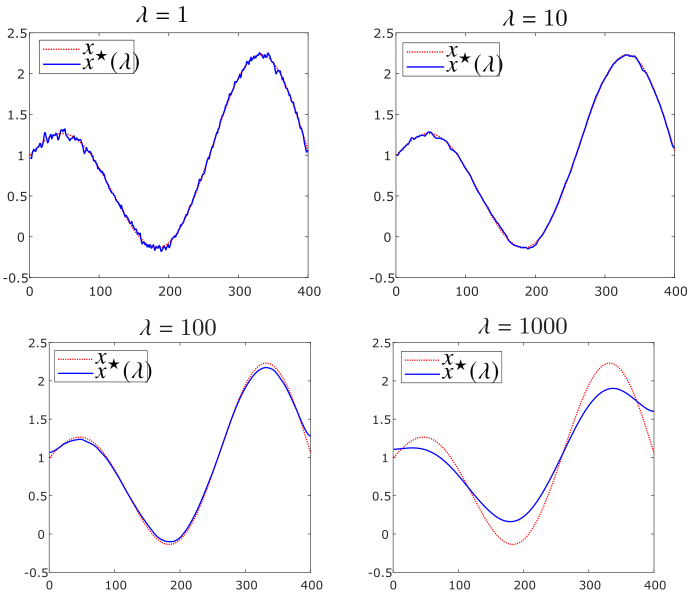
- D is $(n-1) \times n$ difference matrix:

$$D = \begin{bmatrix} -1 & 1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & -1 & 1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -1 & 1 & 0 \\ 0 & 0 & 0 & \cdots & 0 & -1 & 1 \end{bmatrix}$$

- the optimal solution is given by

$$x^\star(\lambda) = (I + \lambda D^T D)^{-1} y$$

Example: signal denoising



Example: least squares image deblurring

$$y = Ax + v$$

- x is unknown image, y is observed blurred noisy image
- A is (known) blurring matrix, v is (unknown) noise
- images are $M \times N$, stored as MN-vectors

Least squares deblurring

$$\text{minimize} \quad \|Ax - y\|^2 + \lambda \left(\|D_\mathrm{h}x\|^2 + \|D_\mathrm{v}x\|^2\right)$$

- 1st term is "data fidelity" term: ensures $A\hat{x} \approx y$
- 2nd term penalizes differences between values at neighboring pixels

$$\|D_\mathrm{h}x\|^2 + \|D_\mathrm{v}x\|^2 = \sum_{i=1}^{M} \sum_{j=1}^{N-1} \left(X_{i,j+1} - X_{ij}\right)^2 + \sum_{i=1}^{M-1} \sum_{j=1}^{N} \left(X_{i+1,j} - X_{ij}\right)^2$$

when X is the $M \times N$ image stored in the MN-vector x

Example: least squares image deblurring

suppose x is the $M \times N$ image X, stored column-wise as MN-vector

$$x = \left(X_{1:M,1}, X_{1:M,2}, \ldots, X_{1:M,N}\right)$$

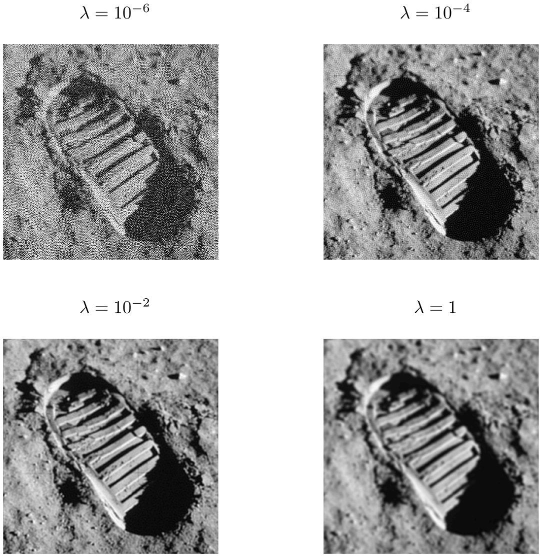
Horizontal differencing: $(N-1) \times N$ *block* matrix with $M \times M$ blocks

$$D_{\mathrm{h}} = \begin{bmatrix} -I & I & 0 & \cdots & 0 & 0 & 0 \\ 0 & -I & I & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & -I & I \end{bmatrix}$$

Vertical differencing: $N \times N$ *block* matrix with $(M-1) \times M$ blocks

$$D_{\mathrm{v}} = \begin{bmatrix} D & 0 & \cdots & 0 \\ 0 & D & \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & \cdots & D \end{bmatrix}, \quad D = \begin{bmatrix} -1 & 1 & 0 & \cdots & 0 & 0 \\ 0 & -1 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -1 & 1 \end{bmatrix}$$

Example: least squares image deblurring



Example: tomography

goal: reconstruct a (density) function $d : \mathbb{R}^2 \to \mathbb{R}$ from line integral measurements

- measurements obtained by passing a beam of radiation through region of interest, and measuring the intensity of the beam after it exits region
- used in medicine, manufacturing, networking, geology
  - common application: CAT (computer-aided tomography) scan

Line integral: parametrize line ℓ in 2-D as

$$p(t) = (x_0, y_0) + t(\cos\theta, \sin\theta)$$

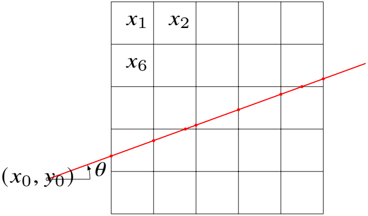
- (x_0, y_0) is any point on the line
- θ is angle measured from horizontal; t is length along line
- line integral of d on ℓ is $\int_\ell d = \int_{-\infty}^{\infty} d(p(t))dt$
- can be extended to 3-D

Line integral measurements

- assume d is constant on pixel (or voxel) i with value x_i
- measurement of integral along line i through region is

$$y_i = \int_{-\infty}^{\infty} d(p_i(t))dt + v_i = \sum_{j=1}^{n} A_{ij}x_j + v_i \quad \text{where } v_i \text{ is small noise}$$

  - $p_i(t)$ is parametrization of line ℓ_i
  - A_{ij} is the length of measurement line i in pixel j
- in matrix-vector form: $y = Ax + v$



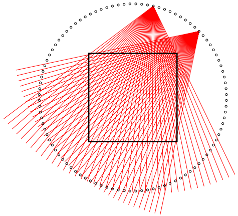
$$\begin{aligned} y = {} & 1.06x_{16} + 0.8x_{17} + 0.27x_{12} + 1.06x_{13} \\ & + 1.06x_{14} + 0.53x_{15} + 0.54x_{10} + v \end{aligned}$$

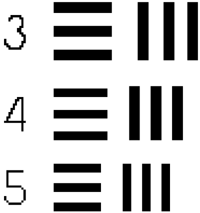
Least squares tomographic reconstruction

$$\text{minimize} \quad \|Ax - y\|^2 + \lambda \left(\|D_\mathrm{h} x\|^2 + \|D_\mathrm{v} x\|^2 \right)$$

D_h and D_v are defined as in image deblurring example (page 9.38)

Example

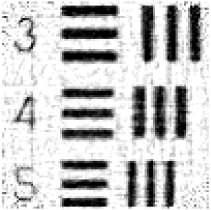




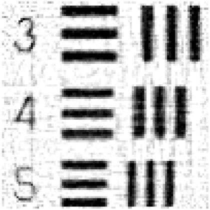
- left: 4000 lines (100 points, 40 lines per point)
- right: object placed in the square region on the left
- region of interest is divided in 10000 pixels

Regularized least squares reconstruction

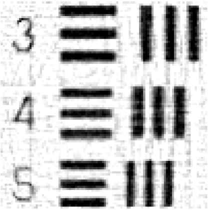
$\lambda = 10^{-2}$



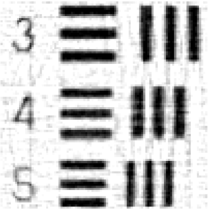
$\lambda = 10^{-1}$



$\lambda = 1$

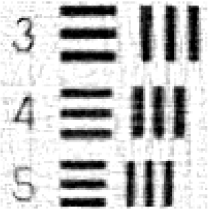


$\lambda = 5$

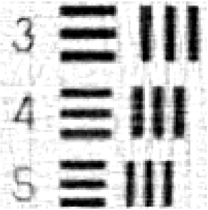


Regularized least squares reconstruction

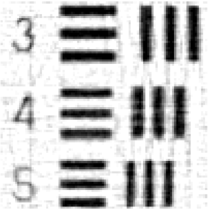
$\lambda = 1$



$\lambda = 5$



$\lambda = 10$



$\lambda = 100$

References and further readings

- S. Boyd and L. Vandenberghe. *Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares.* Cambridge University Press, 2018.

- L. Vandenberghe, *EE133A Lecture Notes,* University of California, Los Angeles.